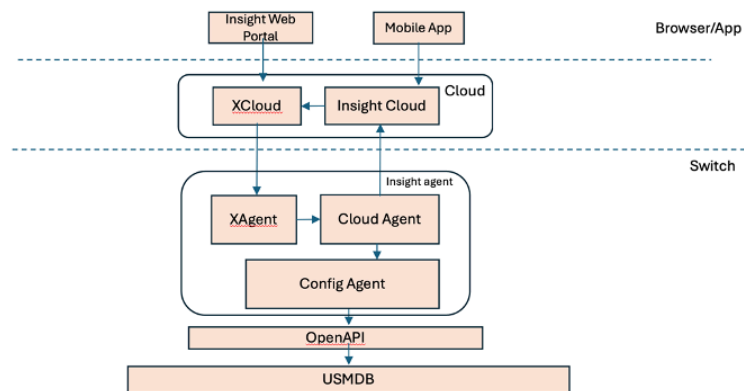
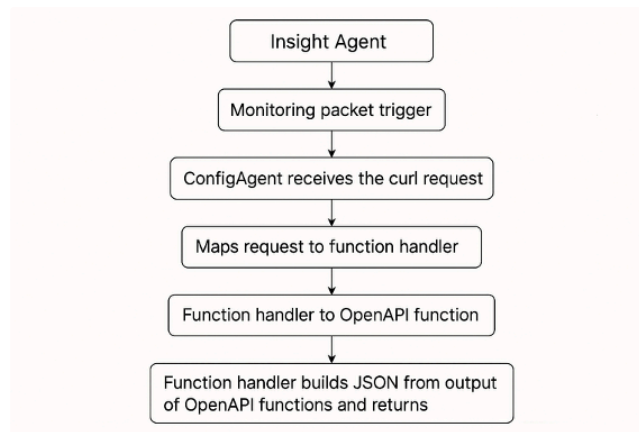


How Insight GET from SWITCH



Insight Monitoring Packet Flow



Monitoring Data

- The **monitoring packet** in Insight acts as a periodic data collection mechanism that gathers critical operational and configuration details from the managed switch.
- The `monitoring_data_thread` retrieves monitoring packets every **10–15 minutes**.
- It calls `mon_packet`, which has all the api.
- `monitoring_data_thread` uses `getApiData` with URL format: <http://127.0.0.1:8444/api/v1/> to fetch data via `curl_get(url)`.

Agent

- The `agent.c` file starts a daemon using `start_mircoshttpd (MHD_start_daemon)`:
 - Listens on **port 8444** and **loopback/any IP**.
- Handles incoming curl requests:
 - If request is **GET**, triggers `create_response`.
 - Processes request via `process_request`.
 - Retrieves handler using `get_uri_entry` from `uri_tbl` (contains API-to-handler mapping).

Example:

API: http://127.0.0.1:8444/api/v1/device_info

Handler: system_device_info

Sample Response

curl -k http://127.0.0.1:8444/api/v1/device_info

```
1 curl -k http://127.0.0.1:8444/api/v1/device_info
2 {
3   "resp": {
4     "status": "success",
5     "respCode": 0,
6     "respMsg": "Operation success"
7   },
8   "deviceInfo": {
9     "serialNumber": "6VED5258A028F",
10    "macAddr": "28:94:01:67:7C:DD",
11    "model": "M4250-12M2XF",
12    "swVer": "13.0.5.20",
13    "numOfPorts": 14,
14    "numOfActivePorts": 4,
15    "memoryUsage": "23.27%",
16    "cpuUsage": "10.31%",
17    "fanState": "Operational",
18    "poeState": false,
19    "upTime": "00 Days 00 Hrs 21 Mins 51 Secs",
20    "temperatureSensors": [
21      {
22        "sensorNum": 1,
23        "sensorDesc": "sensor-1",
24        "sensorTemp": 45,
25        "sensorState": 2
26      }
27    ],
28    "bootVersion": "1.0.0.11",
29    "rxData": 80795,
30    "txData": 483306,
31    "insightAgentVersion": "7.8.0.78000015"
32  }
33 }
34
```

API-to OpEN API Mapping

- **Serial Number:** openapiSystemRegSerialNumGet
- **MAC Address:** openapiSystemMacAddrGet
- **Switch Model:** openapiSystemRegMachineModelGet
- **Version:** openapiNetworkOSVersionGet
- **Number of Ports:** openapiIfFirstGet, openapiIfStatusGet, openapiIfNextGet
- **Active Ports:** openapiIfLinkStateGet
- **Memory & CPU Usage:** sysinfo (*to be verified*)
- **Fan State:** openapiBoxsNumOfFansGet, openapiBoxsFanStateGet
- **PoE Support:** openapiPoeIsSupported
- **Temperature:**
 - openapiBoxsNumOfTempSensorsGet
 - openapiBoxsTempSensorDescripGet
 - openapiBoxsTempGet
 - openapiBoxsTempSensorStatusGet
- **Uptime:** openapiSystemUpTimeGet
- **Boot Version:** openapiBootCodeVersion

- **Rx/Tx Data:** openapiCpuIntfGet
 - OPEN_CTR_RX_TOTAL_BYTES
 - OPEN_CTR_TX_TOTAL_BYTES

APIs called in Monitoring thread:

```
1 • deviceInfo
2 • deviceIpSettings
3 • deviceTime
4 • deviceName
5 • switchStatsPort
6 • connectedClients
7 • poePortConfig
8 • switchConfigLagGroup
9 • uplinkPort
10 • spanningTree
11 • syslogLogging
12 • ipAclList, macAclList, ipAcl, macAcl
13 • switchConfigVlan, privateVlan, vlanMembership
14 • devicePortMirroring
15 • switchPortConfig
16 • l2LoopConfig
17 • radiusStatus, radiusPortConfig, radiusConfig
18 • snmpTrapStatus, trapRecipientIPAddress, snmpPublicCommunity
19 • staticRouting, vlanRouting, vlanIpConfiguration
20 • dhcpL2Relay, dhcpL2RelayPort
21 • dhcpSnoopingGlobal, dhcpSnoopingInterface
22 • protectedPortConfig
23 • poeSchedule, advancePoeConfig
24 • secureDiagnosticsMode
25 • autoTrunk, autoLag
26
```